Table 1: Benchmark summary: mean over all datasets (95% CI beneath, smaller font) for each metric and generator. **Bold** = best per row. Arrows give the preferred direction; MIA AUROC is best near 0.5.

	Metric	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals
Statistical fidelity / Downstream utility (TSTR)	F1-macro ↑	**0.641** ±**0.059**	0.468 ±0.058	0.404 ±0.123	0.540 ±0.055	0.598 ±0.071	0.618 ±0.045	0.424 ±0.039	0.573 ±0.048	0.535 ±0.055	0.410 ±0.045
	F1-weighted ↑	0.681 ±0.064	0.550 ±0.074	0.458 ±0.134	0.639 ±0.061	0.681 ±0.064	**0.686** ±**0.052**	0.532 ±0.068	0.663 ±0.052	0.630 ±0.063	0.458 ±0.056
	Accuracy ↑	0.687 ±0.061	0.582 ±0.076	0.494 ±0.136	0.672 ±0.063	**0.713** ±**0.059**	0.705 ±0.050	0.587 ±0.072	0.689 ±0.053	0.666 ±0.059	0.467 ±0.054
	JS div. ↓	0.064 ±0.022	0.085 ±0.031	0.133 ±0.025	0.045 ±0.027	0.081 ±0.055	0.067 ±0.022	0.170 ±0.050	0.225 ±0.067	**0.034** ±**0.012**	0.205 ±0.043
	KS stat. ↓	0.166 ±0.037	0.212 ±0.042	0.272 ±0.036	**0.105** ±**0.034**	0.117 ±0.053	0.180 ±0.030	0.346 ±0.053	0.243 ±0.041	0.112 ±0.015	0.388 ±0.059
	Hellinger ↓	0.175 ±0.027	0.192 ±0.045	0.303 ±0.034	**0.122** ±**0.037**	0.167 ±0.072	0.179 ±0.037	0.293 ±0.053	0.338 ±0.080	0.134 ±0.022	0.354 ±0.052
	Corr. pres. ↑	0.910 ±0.028	0.824 ±0.053	0.847 ±0.033	**0.930** ±**0.015**	0.887 ±0.031	0.914 ±0.018	0.801 ±0.052	0.868 ±0.039	0.864 ±0.037	0.821 ±0.053
Privacy / disclosure	DCR ↑	0.342 ±0.074	0.552 ±0.129	0.258 ±0.069	0.400 ±0.090	0.747 ±0.498	0.359 ±0.082	0.605 ±0.158	0.337 ±0.057	0.405 ±0.096	**1.388** ±**0.391**
	NNDR ↑	0.888 ±0.037	0.876 ±0.042	0.951 ±0.016	0.851 ±0.051	0.821 ±0.070	0.874 ±0.042	0.885 ±0.035	**0.952** ±**0.021**	0.861 ±0.046	0.872 ±0.048
	MIA AUROC →.5	0.545 ±0.020	0.503 ±0.008	0.502 ±0.007	**0.501** ±**0.007**	0.537 ±0.024	0.511 ±0.007	0.499 ±0.004	0.513 ±0.010	0.503 ±0.008	0.502 ±0.006
	Dup. rate ↓	0.043 ±0.047	**0.002** ±**0.003**	0.028 ±0.039	0.006 ±0.008	0.059 ±0.085	0.019 ±0.031	0.008 ±0.014	0.018 ±0.028	0.015 ±0.026	0.003 ±0.005
Diversity / coverage (PRDC)	Precision ↑	0.879 ±0.082	0.619 ±0.118	**0.900** ±**0.087**	0.824 ±0.090	0.816 ±0.111	0.821 ±0.097	0.517 ±0.116	0.824 ±0.083	0.728 ±0.127	0.427 ±0.120
	Recall ↑	0.732 ±0.111	0.895 ±0.038	0.311 ±0.088	0.929 ±0.031	**0.964** ±**0.020**	0.857 ±0.031	0.838 ±0.077	0.623 ±0.103	0.940 ±0.022	0.921 ±0.072
	Density ↑	0.952 ±0.173	0.414 ±0.104	**1.187** ±**0.235**	0.677 ±0.114	0.943 ±0.243	0.810 ±0.189	0.343 ±0.140	0.950 ±0.211	0.671 ±0.190	0.219 ±0.098
	Coverage ↑	0.758 ±0.103	0.532 ±0.110	0.392 ±0.079	0.775 ±0.105	**0.881** ±**0.106**	0.718 ±0.093	0.367 ±0.091	0.637 ±0.107	0.705 ±0.128	0.216 ±0.104
Detection	C2ST ↓	0.894 ±0.056	0.934 ±0.033	0.961 ±0.021	0.817 ±0.067	**0.724** ±**0.085**	0.908 ±0.045	0.967 ±0.020	0.979 ±0.030	0.875 ±0.058	0.973 ±0.018
	MMD ↓	0.041 ±0.013	0.097 ±0.032	0.181 ±0.041	0.020 ±0.017	0.070 ±0.065	0.039 ±0.012	0.232 ±0.074	0.046 ±0.021	**0.018** ±**0.004**	0.387 ±0.090
	Wasserstein ↓	0.230 ±0.036	0.406 ±0.085	0.453 ±0.059	**0.168** ±**0.067**	0.558 ±0.390	0.260 ±0.037	0.656 ±0.113	0.309 ±0.053	0.178 ±0.014	1.728 ±0.649
SDMetrics	Quality ↑	0.630 ±0.077	0.573 ±0.065	0.551 ±0.071	**0.676** ±**0.077**	0.676 ±0.099	0.642 ±0.081	0.505 ±0.064	0.567 ±0.074	0.634 ±0.072	0.468 ±0.055
	Col. shapes ↑	0.770 ±0.030	0.737 ±0.045	0.673 ±0.036	**0.834** ±**0.025**	0.793 ±0.050	0.771 ±0.039	0.634 ±0.054	0.724 ±0.049	0.815 ±0.026	0.587 ±0.047
	Pair trends ↑	0.490 ±0.147	0.410 ±0.117	0.429 ±0.130	0.518 ±0.149	**0.558** ±**0.181**	0.514 ±0.152	0.377 ±0.109	0.411 ±0.133	0.454 ±0.131	0.348 ±0.104

Table 2: DISCO-TAB vs. the single best competing generator per metric (mean over all datasets ±95% CI). **Bold** marks the winner.

	Metric	DISCO-TAB	Best competitor	(method)
Statistical fidelity — Downstream utility (TSTR)	F1-macro ↑	**0.641 ±0.059**	0.618 ±0.045	ADS-GAN
	F1-weighted ↑	0.681 ±0.064	**0.686 ±0.052**	**ADS-GAN**
	Accuracy ↑	0.687 ±0.061	**0.713 ±0.059**	**TabDDPM**
Statistical fidelity	JS div. ↓	0.064 ±0.022	**0.034 ±0.012**	**NFlow**
	KS stat. ↓	0.166 ±0.037	**0.105 ±0.034**	**GaussCop**
	Hellinger ↓	0.175 ±0.027	**0.122 ±0.037**	**GaussCop**
	Corr. pres. ↑	0.910 ±0.028	**0.930 ±0.015**	**GaussCop**
Privacy / disclosure	DCR ↑	0.342 ±0.074	**1.388 ±0.391**	**Marginals**
	NNDR ↑	0.888 ±0.037	**0.952 ±0.021**	**RTVAE**
	MIA AUROC →.5	0.545 ±0.020	**0.501 ±0.007**	**GaussCop**
	Dup. rate ↓	0.043 ±0.047	**0.002 ±0.003**	**CTGAN**
Diversity / coverage (PRDC)	Precision ↑	0.879 ±0.082	**0.900 ±0.087**	**TVAE**
	Recall ↑	0.732 ±0.111	**0.964 ±0.020**	**TabDDPM**
	Density ↑	0.952 ±0.173	**1.187 ±0.235**	**TVAE**
	Coverage ↑	0.758 ±0.103	**0.881 ±0.106**	**TabDDPM**
Detection	C2ST ↓	0.894 ±0.056	**0.724 ±0.085**	**TabDDPM**
	MMD ↓	0.041 ±0.013	**0.018 ±0.004**	**NFlow**
	Wasserstein ↓	0.230 ±0.036	**0.168 ±0.067**	**GaussCop**
SDMetrics	Quality ↑	0.630 ±0.077	**0.676 ±0.077**	**GaussCop**
	Col. shapes ↑	0.770 ±0.030	**0.834 ±0.025**	**GaussCop**
	Pair trends ↑	0.490 ±0.147	**0.558 ±0.181**	**TabDDPM**